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Recovery-Aware Physical Attribution of Load and Line Events From Sparse PMUs

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ABSTRACT After a load is restored or a line is reclosed, residual dynamics can resemble a new intervention. We study sequential attribution from eight PMUs in an IEEE 39-bus model with nine candidate load buses and nine lines. A bank of switched differential-algebraic predictors compares intervention and recovery histories from estimated past states. Conditional Bayesian evidence integrates load severity, while separate active and cause weights retain the explanation of a restored intervention. In an independent simulation evaluation, 108 interventions are replayed under three network conditions and two noise levels. The physical bank produces 644/648 correct family/source decisions before restoration with 72 unmatched alarms, versus 330/648 with 29 for a fixed causal ExtraTrees adapter. The gain concentrates on loads; line coverage matches the short-window learned comparator. A controlled posthoc study identifies a state-carry benefit under both marginal and profiled severity scoring: retaining the state gives 108/108 timely decisions and one extra alarm, compared with 75/108 and 166 after resetting physical deviations. The results establish recovery-consistent prediction as a useful design choice for sparse-PMU attribution and quantify its coverage–alarm trade-off under network uncertainty.

INDEX TERMS event detection, model uncertainty, phasor measurement units, physics-informed inference, sequential diagnosis, source localization.

I. INTRODUCTION

AFTER a load is restored or a line is reclosed, the machines continue moving. A second intervention may begin before that response has decayed. With only eight PMUs in a 39-bus network, a change in a measured signal may therefore indicate a new source, recovery from an earlier source, or an imperfect physical prediction. The task is to distinguish these explanations using only the reports available at the decision time.

Physical restoration is not electrical recovery. A detector that returns immediately to a flat reference can issue another alarm for the remaining oscillation. Suppressing alarms throughout recovery can instead hide a real intervention. The operating point adds another difficulty: correcting a voltage or current offset need not correct how the network responds to a later input.

Existing work provides strong foundations. DAE state estimators infer unmeasured electrical states [1]; physical outage detectors compare network hypotheses sequentially [2], [3]; and simulated fault trajectories support diagnosis from sparse PMUs [4]. Physical models, transient histories and unknown-onset searches are thus established ingredients. A small state-reconstruction error, however, does not by itself identify the intervention that produced the observations.

We compare physical histories rather than isolated signal features. A history specifies an intervention, source, onset, severity and restoration. DAE-derived predictors start from estimated past states and preserve the state left by previous interventions. Conditional evidence integrates load severity, accounts for instrumental memory, and updates fixed network-model hypotheses. Separate active and cause masses keep a restored intervention available as an explanation of the continuing response.

The paper makes three contributions within this restricted problem:

1) An executable attribution mechanism that makes new load/line interventions compete with retained recovery histories. We specify its physical predictors, evidence updates and decision rules.
2) A comparison that separates detection, correct source

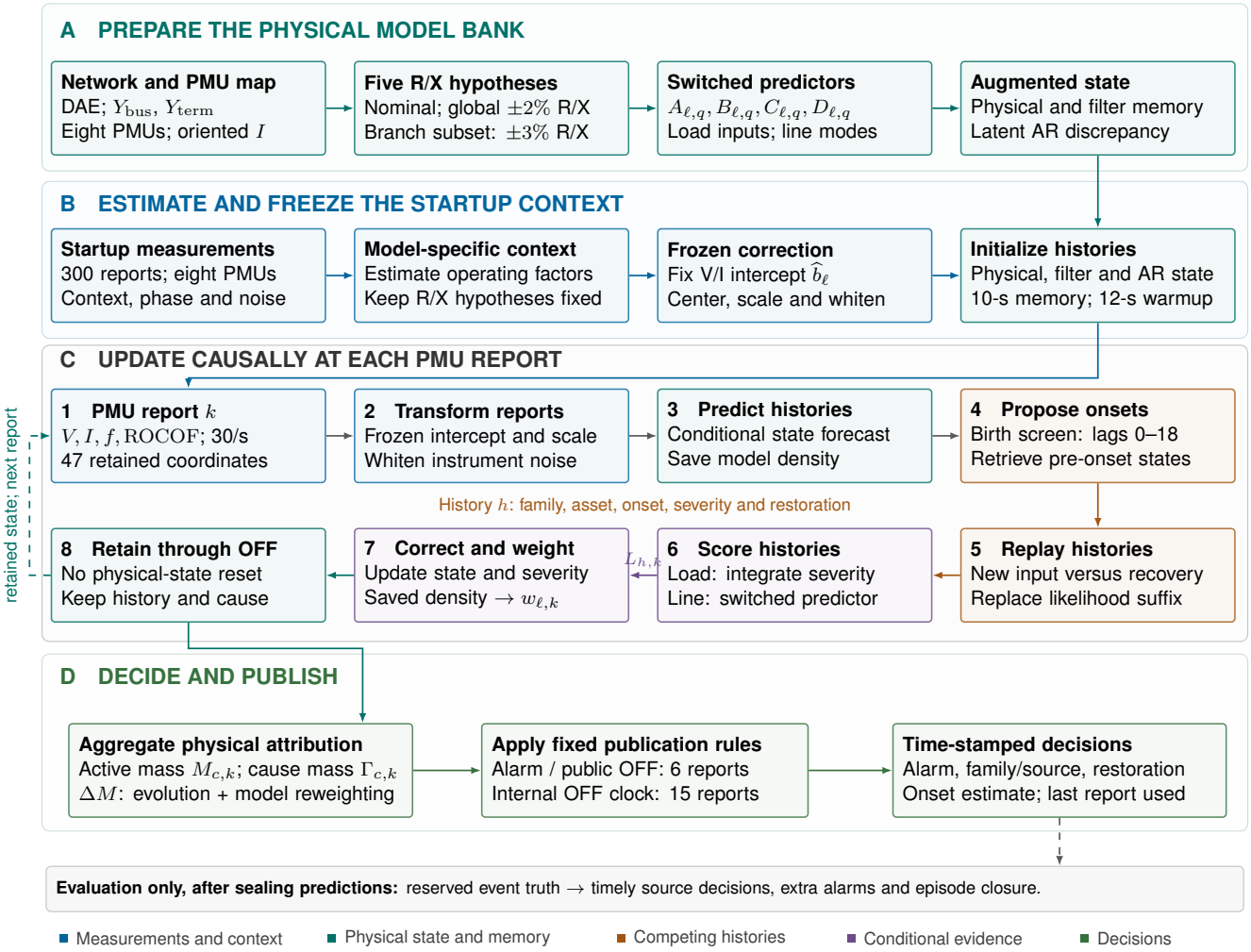


FIGURE 1. Implemented physical-history attribution, organized into model preparation (A), startup context (B), per-report inference (C), and publication (D). The bank predicts before state correction and saves model evidence before proposal selection. Candidate replay replaces a likelihood suffix, so each report contributes once. The feedback loop retains physical and instrumental memory through restoration. Reserved event truth is used only after predictions are sealed.

attribution before restoration, extra alarms and episode closure. Nominal and uncertain network models are evaluated alongside two fixed ExtraTrees adapters; Chen's published detector is compared on eligible persistent line openings.

3) A controlled severity-score/state-carry study on identical records. It quantifies the benefit of preserving physical-state deviations at inferred closure under both marginalization and bounded profiling. The common observations and operators connect the decision improvement to a specific state-retention operation.

The independent comparison gives 644/648 timely attributions with 72 extra alarms. The advantage over the fixed learned adapters concerns load coverage, while network mismatch increases alarm burden. This evaluates configured procedures at their original thresholds. It motivates recovery-consistent inference without establishing superiority at equal false-alarm rates. Auxiliary voltage, feature, simultaneous-source and context-signature studies are retained in the supplement with their own cohorts.

II. RELATED WORK

A. STATE ESTIMATION AND PHYSICAL SOURCE INFERENCE

Unknown-input and DAE estimators recover electrical states despite incomplete models or exogenous inputs [1], [5], [6]. Functional observability asks whether a requested state quantity can be recovered [7], [8]. Source inference asks which intervention can explain the trajectory. Descriptor-model diagnosability and nuisance-profiled tests formalize this distinction [9], [10].

Physical hypotheses support outage inference from phasor angles [11], Bayesian inference with uncertain states [12], and sequential detection under stochastic operation [2], [3]. Anguluri *et al.* recover sparse forcing locations, inputs and an unknown initial state using group LASSO for linear dynamics

[13]. These are precedents for source recovery; group LASSO is not an executed comparator in our frozen campaign. Sensor placement also affects source distinguishability [14]–[16].

Recent learned reconstruction also targets sparse sensing. Pomarico *et al.*'s SHRED preprint evaluates dynamic-state reconstruction on IEEE-39 from a limited number of PMUs [17]. Its target is state reconstruction rather than the source/ON/OFF episode contract here. Dey *et al.* detect and localize distribution-network topology changes from phasors using learned statistics, with hardware-in-the-loop validation [18]. Neither sparse sensing nor change localization alone distinguishes the present contribution. These are adjacent tasks, not executed performance baselines in this study.

B. TRAJECTORY-BASED AND SEQUENTIAL DIAGNOSIS

Warner *et al.* compare simulated voltage/current fault trajectories over a moving onset window and return candidate lines [4]. Their evaluated target is short-circuit identification before clearing; load changes are an adverse-event test. Our target follows load changes and line openings through restoration and successive interventions. Physical trajectory libraries, remote PMUs and retrospective onset scanning are shared ingredients. Their procedure was not run on our data, so no numerical superiority is inferred.

The Adaptive Matrix CuSum derives sequential identification error/delay results under specified observation laws [19]. Multiple-model filtering already propagates conditional state/mode hypotheses [20]. The multiple-model multiple-hypothesis filter maintains mode-sequence histories and adapts their number through pruning and merging [21]. Retaining histories or adapting bank size is therefore not an original principle of our method. Our pruned history bank uses these ideas, but its approximate covariances, proposal screen and fitted startup context do not inherit their general guarantees. The complete procedure evaluated here makes retained recovery compete with later interventions and reports timely attribution and alarm burden. These comparisons do not isolate recovery memory from the other admission and publication rules.

C. LEARNED DIAGNOSIS AND PHYSICAL UNCERTAINTY

Learned and knowledge-driven representations identify PMU disturbances and assets [22]–[24]. We compare two fixed ExtraTrees adapters from the SGSMA project. Their performance tests transfer of those implementations, not the limits of all supervised learning. A physical catalog and a fixed learned adapter have different access to candidate-source responses. This comparison does not equalize that information or establish a source-excluded generalization result.

Physical stochastic models support latent-state estimation [25], [26], while joint DAE parameter inference addresses uncertain generators and networks [27]. Our stream method uses a finite bank of constant network hypotheses. Trajectory sensitivities and Gaussian evidence have also been combined for Bayesian parameter inference in power-system DAEs [28]. Their use alone does not distinguish an event attribution algorithm. The separate signature test examines a local event–context derivative; it does not identify every plant parameter or replace the stream method with an unrestricted joint estimator.

III. PROBLEM AND MEASUREMENT MODEL

At report k, inference uses measurements through $t_k = k/30$ to decide whether a physical intervention occurred. Detection concerns its presence; classification selects a physical operator; localization selects a bus or branch. These outputs can become reliable at different times. We record the estimated physical onset $\widehat{\tau}$ separately from the alarm and diagnosis emission times.

Figure 2 introduces the broader measurement phenomena using the known RAW0001 reference record. Faults, line changes, generation changes and load changes alter the physical network; missing or corrupted reports alter observation integrity. The present causal test addresses load changes and line openings with restoration. The other panels provide context, not additional validated event classes.

The network has 39 buses, but the prescribed PMU set is

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}. \tag{1}$$

Voltage is measured at each PMU bus. Current is measured at a specified branch terminal, which can differ from that bus. The causal implementation uses 47 coordinates: eight voltage magnitudes, eight voltage angles, eight current magnitudes, seven current angles, eight absolute frequencies and eight ROCOFs, at 30 reports/s. BUS39 current angle is excluded because its interpretation remains partially validated. The BUS5 current terminal retains the provisional historical 5–6 assignment. These are declared measurement assumptions, not identities inferred from classification accuracy.

The exported magnitudes are in V and A, angles in degrees, frequency in Hz and ROCOF in Hz/s. Internal centered coordinates do not replace the absolute frequency contract. These are reporting-rate phasor measurements; waveform harmonics or THD are not measured by these channels.

The causal stream is exported from one balanced complex phasor per PMU: phase A supplies the voltage and current magnitudes and angles. The CSV exporter constructs B and C by equal magnitudes and angle offsets of -120° and $+120^\circ$. Retaining A therefore loses no independent phase information in these synthetic records; it is not measured positive-sequence extraction or a model of unbalanced faults. The vector is $y_k = (|V_A|_{\mathcal{P}}, \angle V_{A\mathcal{P}}, |I_A|_{\mathcal{P}}, \angle I_{A\mathcal{P}\setminus\{39\}}, f_{\mathcal{P}}, \dot{f}_{\mathcal{P}})_k$. The supplementary measurement table gives all eight current-terminal assignments, polarities, gains and phase offsets. Products of voltage and current measured at different terminals are power proxies.

The physical model is

$$\dot{x} = f_q(x, z, u; \theta), \qquad 0 = g_q(x, z, u; \theta), \tag{2}$$

$$\begin{aligned} h_k &= H_q(x_k, z_k, u_k), \\ \eta_k &= F_M(\eta_{k-1}, h_k; \theta_M), \\ y_k &= G_M(\eta_k, h_k; \theta_M) + b + \varepsilon_k. \end{aligned} \tag{3}$$

TABLE 1. Closest inference contracts. Only Chen and the fixed ExtraTrees adapters are executed comparators in the confirmatory stream campaign. An untested capability is not an algorithmic impossibility.

Route	Mechanism	Evaluated target	Role in this study
Chen / Rovatsos	Network likelihoods and sequential tests	Line outages; Rovatsos includes transients	Chen on persistent-line first decisions
Warner high-fidelity	Simulated V/I paths and onset search	Short-circuit line candidates	Closest trajectory precedent; no common-data run
Adaptive Matrix CuSum	Sequential diagnosis with error/delay analysis	One unknown distributional change	Statistical precedent; no inherited guarantee
IMM / M3H	Conditional filters, mode histories and mixture reduction	Hybrid switching-state estimation	Direct history-filter precedent; no common-data baseline here
DAE / stochastic-state estimation	Physical state and uncertainty models	States, unknown inputs or parameters	Components and adjacent tasks
Learned PMU diagnosis	Features or constrained representations	Event families and locations	Two frozen ExtraTrees references
Physical history bank here	Estimated-past-state paths and recovery	Successive catalog load/line events	Timely attribution, extras and restoration

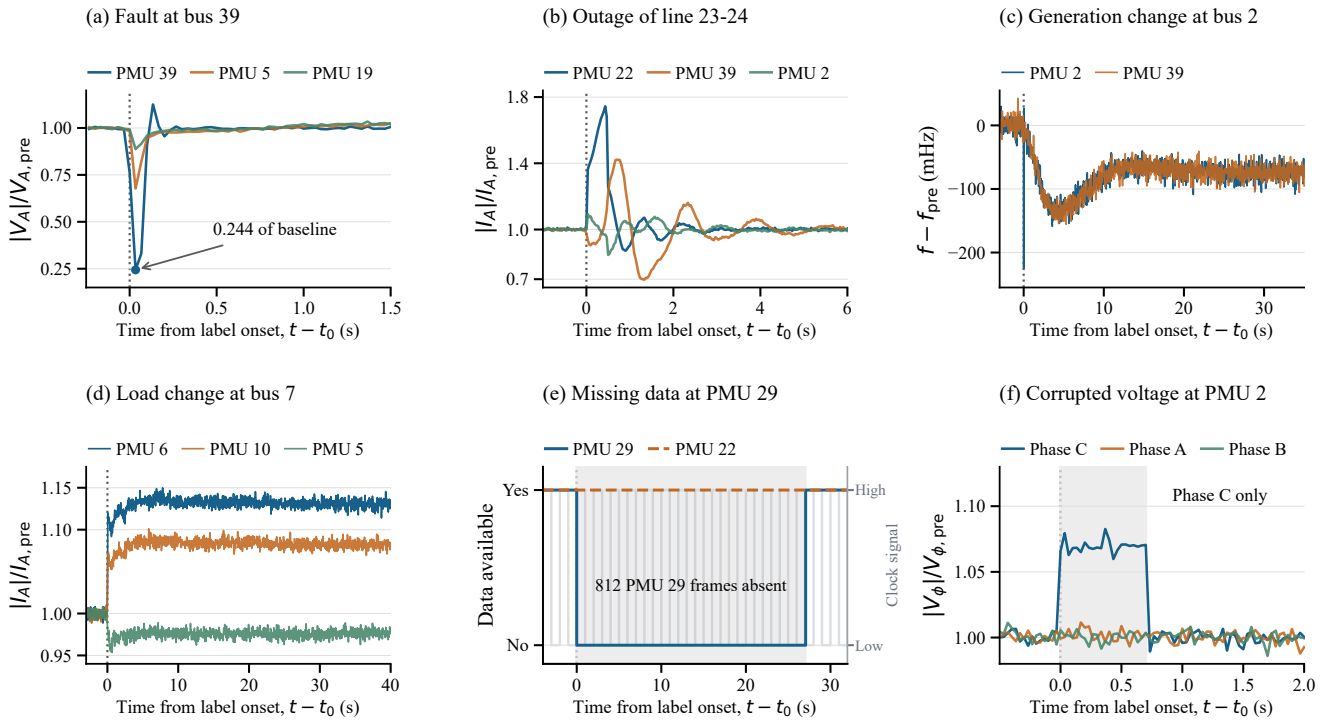


FIGURE 2. Representative signatures from the archived RAW0001 reference simulation: fault, line outage, generation change, load change, missing PMU data and corrupted voltage. Alignment uses retrospective label onset for illustration; inference receives no such alignment. The gray clock in (e) is an illustrative reference, not another measured channel. Only load and line mechanisms enter the causal evaluation below. RAW0001 is known retrospective evidence, not independent field validation.

Here x contains machine and control dynamics, z the algebraic network variables, q the topology/intervention mode, and θ_M the measurement convention. H_q selects electrical phasors; η_k retains the causal phase history used for frequency/ROCOF and the measurement-filter state. The observation thus depends on an augmented state. Bus frequency comes from phase evolution, not a machine's rotor speed. Instrumental noise is applied according to the declared generator contract.

An intervention history is $h = (e, c, \tau, a, \kappa)$: family, source, onset, severity and restoration time, with $\kappa = \infty$ for a sustained event. Its duration is $\kappa - \tau$ when restoration occurs. A load changes its local P/Q setpoints; a line opening switches its branch primitive. For the supported impulse-free switches,

$$x(\tau^+) = x(\tau^-), \qquad g_{q^+}(x(\tau^-), z(\tau^+), u; \theta) = 0. \quad (4)$$

The algebraic voltage/current response may jump even though dynamic states remain continuous. Physical restoration reverses the intervention operator. It does not return the machines to their pre-event state, so electrical recovery can continue after OFF.

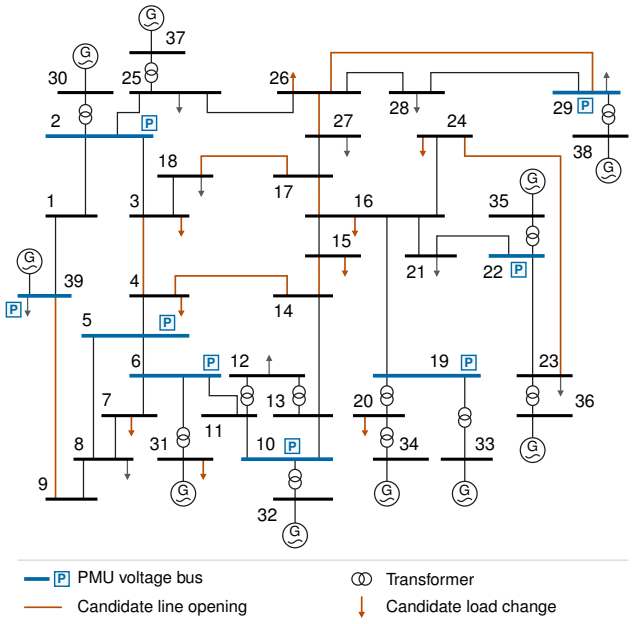


FIGURE 3. **IEEE 39-bus single-line diagram: 46 branches, 12 transformers and 10 generators. Blue bars and P marks locate the eight voltage PMUs. Orange branches and load arrows identify the nine line and nine load candidates; gray arrows denote other loads. Current terminals follow the separate measurement contract. All routes are orthogonal.**

TABLE 2. **Scope of the executed causal component. The source is an intervention asset; it is not necessarily a measured PMU bus.**

Case	Physical operator	Executed claim
Normal	No intervention	Monitor after startup
Load	Local P/Q factor; optional restoration	Nine candidate buses
Line	Branch open; optional reclose	Nine candidate branches
Fault/generation	Different shunt/input/trip operators	Outside this causal test
Missing/bad data	Measurement-integrity mechanism	Outside the stream test
Unknown	Unresolved diagnosis	Not a new physical family

Switch-aligned reports use the consistent post-switch algebraic solution. The local reduction requires an invertible algebraic Jacobian after fixing the angle reference; the catalog excludes islanding line alternatives. Successful solves and local residual checks validate the admitted cases, not global regularity for arbitrary switching topologies.

The generator stores event schedules and true states separately from the PMU stream. Inference receives the nominal network, permitted calibration and reports through the current time. It receives no source, switch time, true R/X realization or solver state. Initial operating context and noise scale are estimated from 300 startup reports; monitoring begins after the inherited 12-s warmup. The startup contract assumes no intervention in that prefix.

IV. RECOVERY-AWARE PHYSICAL ATTRIBUTION

Figure 1 describes the implemented computation. The inference engine uses a conditional history bank of DAE-derived switched state-space predictors. The nonlinear simulator generates observations; inference does not solve a new full nonlinear simulation for every candidate at every report.

Physics enters the candidate dynamics, input directions and oriented measurement map. Statistical filtering represents uncertain states and discrepancy around those predictions. Thus the evaluated method is model-based inference; it does not train a neural network with a physics loss or learn a source-label classifier. A good residual fit alone is insufficient: competing histories must also yield a timely source decision.

A. CONSTRUCT THE PREDICTOR FROM THE NETWORK

Kirchhoff constraints and complex power connect sources to remote PMUs:

$$I_{bus} = Y_{bus}V, \quad I_{term,r} = Y_{term,r}V, \quad S_i = V_i I^*_{bus,i}. \tag{5}$$

Terminal rows retain orientation, line charging and transformer ratios. A measured terminal current is not a nodal injection. A line opening changes its branch primitive and terminal map; a load changes its admitted P/Q input direction.

On a regular index-one DAE chart with a fixed angle reference,

$$\begin{aligned} \delta z &= -g_z^{-1}(g_x \delta x + g_u \delta u), \\ \delta \dot{x} &= A_q^{red}\delta x + B_q^{red}\delta u, \\ A_q^{red} &= f_x - f_z g_z^{-1} g_x, \quad B_q^{red} = f_u - f_z g_z^{-1} g_u. \end{aligned} \tag{6}$$

For the instantaneous electrical observation map $H_q(x, z, u)$, the corresponding output operators are

$$C_q^{red} = H_x - H_z g_z^{-1} g_x, \qquad D_q^{red} = H_u - H_z g_z^{-1} g_u. \tag{7}$$

They propagate a perturbation at an unmeasured source to the selected voltage and oriented-current channels. Network connectivity enters through the branch stamps in g_q and H_q; eliminating unmeasured algebraic variables generally produces dense reduced operators. The graph therefore constrains the prediction without a learned graph embedding. A changed terminal operator also contributes directly through H_q, not only through the changed voltage. The implementation discretizes the admitted switch modes and adds causal phase-history operators for frequency and ROCOF. Inertia, controls and load law remain those of the fixed plant.

After augmenting instrumental memory, a conditional predictor has the local form

$$\begin{aligned} \xi_{k+1} &= A_{\ell,q}\xi_k + b_{\ell,q} + B_{\ell,q}a + w_k, \\ r_k &= C_{\ell,q}\xi_k + d_{\ell,q} + D_{\ell,q}a + v_k. \end{aligned} \tag{8}$$

Here r_k contains reference-corrected, scaled and instrument-whitened PMU coordinates, ℓ is a fixed network hypothesis, and ξ_k contains physical deviation states, phase history and predicted measurement-filter memory. Observed

colored-instrument filters whiten measurements; their corresponding predictor memory is deterministic. Additional latent AR states represent persistent discrepancy, with preset quiet/load/line/recovery covariance and drift regimes. This statistical discrepancy model is distinct from the plant equations. v_k denotes the remaining innovation. A load has scalar severity within its specified P/Q direction. A line uses a switched operator rather than a fictitious load-amplitude fit. States from different equilibria are not interpreted as an absolute rotor-state mixture.

B. ESTIMATE CONTEXT FROM THE STARTUP PREFIX

Each network hypothesis uses the same 300 startup reports to estimate operating factors, reference phase and noise scale. Voltage/current coordinates receive a constant discrepancy correction,

$$\widehat{b}_\ell = \bar{y}_{0:299} - y_\ell^{ref}, \qquad b_{\ell,k} = \widehat{b}_\ell \quad (k \geq 300). \tag{9}$$

Angular differences respect wrapping and the common reference. Frequency and ROCOF use empirical prefix centers. The correction remains fixed so it cannot track a later load step and subtract its signature. This estimates an observation discrepancy, not the true impedances. Prefix estimates enter by substitution rather than through a full joint parameter posterior.

C. COMPARE UNKNOWN-ONSET HISTORIES

The method retains ten seconds of past conditional states. Its frozen birth screen searches lags of 0–18 reports and proposes selected onsets; it does not enumerate every onset in the 300-report history. Each history specifies family, asset, onset, restoration mode and severity. It starts from the stored conditional state before its proposed onset. That state already contains any ongoing recovery; an alarm does not reset the predictor to equilibrium.

For a retained load history, conditional Gaussian filtering yields a residual e, severity response s and covariance S. Unknown severity has a truncated-Gaussian distribution inherited from previous reports:

$$L_{h,k} = \int_l^u \mathcal{N}(e; sa, S)\, \mathrm{TN}_{[l,u]}(a; \mu, v)\, da. \tag{10}$$

With Φ denoting the standard normal CDF, define

$$\begin{aligned} v' &= (v^{-1} + s^\top S^{-1} s)^{-1}, \\ \mu' &= v'(\mu/v + s^\top S^{-1} e), \\ Z(\mu, v) &= \Phi((u-\mu)/\sqrt{v}) - \Phi((l-\mu)/\sqrt{v}). \end{aligned} \tag{11}$$

The implemented local integral follows by completing the square:

$$\begin{aligned} \log L_{h,k} = {} & \log \mathcal{N}(e; 0, S) + \tfrac{1}{2}\log(v'/v) \\ & + \tfrac{1}{2}(\mu'^2/v' - \mu^2/v) + \log \frac{Z(\mu', v')}{Z(\mu, v)}. \end{aligned} \tag{12}$$

These standard identities are exact for the stated local conditional model. Adaptive proposals, pruning and covariance approximations make the complete bank approximate.

The conditional state mean is represented as $m_h + d_h a$. To make the filtering approximation explicit, let $T_o, J_o, g_o, u_o, j_o, v_o$ denote the archived one-report transition and observation operators for switch o. The code forms $e = r_k - J_o m_h - j_o$ and $s = J_o d_h + v_o$. With the shared covariance P_ρ for the current discrepancy regime ρ,

$$\begin{aligned} S_o &= R + J_o P_\rho J_o^\top, & K_o &= T_o P_\rho J_o^\top S_o^{-1}, \\ m_h^+ &= T_o m_h + g_o + K_o e, & d_h^+ &= T_o d_h + u_o - K_o s. \end{aligned} \tag{13}$$

The amplitude distribution is updated by the same integral above. The regime covariances are obtained from nominal-operator Riccati equations; The implementation does not propagate an independent exact covariance for every switched history. At episode closure, it retains the mixture mean of the augmented state and records the omitted between-history spread. Thus state continuity and conditional Bayesian amplitude integration are implemented, while a fully Bayesian posterior over states, context and all histories is not.

The rolling screen proposes histories; it is not another independent observation. Replaying a suffix replaces its original no-new-event suffix likelihood. The sequential update therefore counts each report once, rather than repeatedly accumulating overlapping ten-second windows.

D. REPRESENT FIXED NETWORK UNCERTAINTY

The bank contains nominal, global $\pm 2\%$, and selected-branch $\pm 3\%$ paired R/X hypotheses. Parameters stay constant within a hypothesis. Its outer weight uses a predictive density formed before the current-report state update and proposal selection:

$$w_{\ell,k} = \frac{w_{\ell,k-1} p_\ell(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_\ell)}{\sum_j w_{j,k-1} p_j(y_k \mid \mathcal{F}_{k-1}, \widehat{b}_j)}. \tag{14}$$

Startup-fitting reports are not scored again as independent model-selection evidence. The mixture covers a finite set of network alternatives rather than identifying a continuous impedance posterior.

E. KEEP RECOVERY AS A COMPETING PHYSICAL EXPLANATION

Restoring an input or topology at κ does not restore the pre-event state: $x(\kappa^+) = x(\kappa^-)$. On a regular local post-restoration chart, the difference from a same-context no-new-intervention trajectory obeys

$$\begin{aligned} \delta x_h(t) &\simeq \Phi_0(t, \kappa)\delta x_h(\kappa), \\ \delta y_h^{net}(t) &\simeq C_0(t)\Phi_0(t, \kappa)\delta x_h(\kappa), \quad t > \kappa. \end{aligned} \tag{15}$$

Here y^{net} denotes voltage/current coordinates before the instrument map. Instrumental memory is retained as well. This local interpretation of known dynamics explains why the reference after OFF is a predicted trajectory.

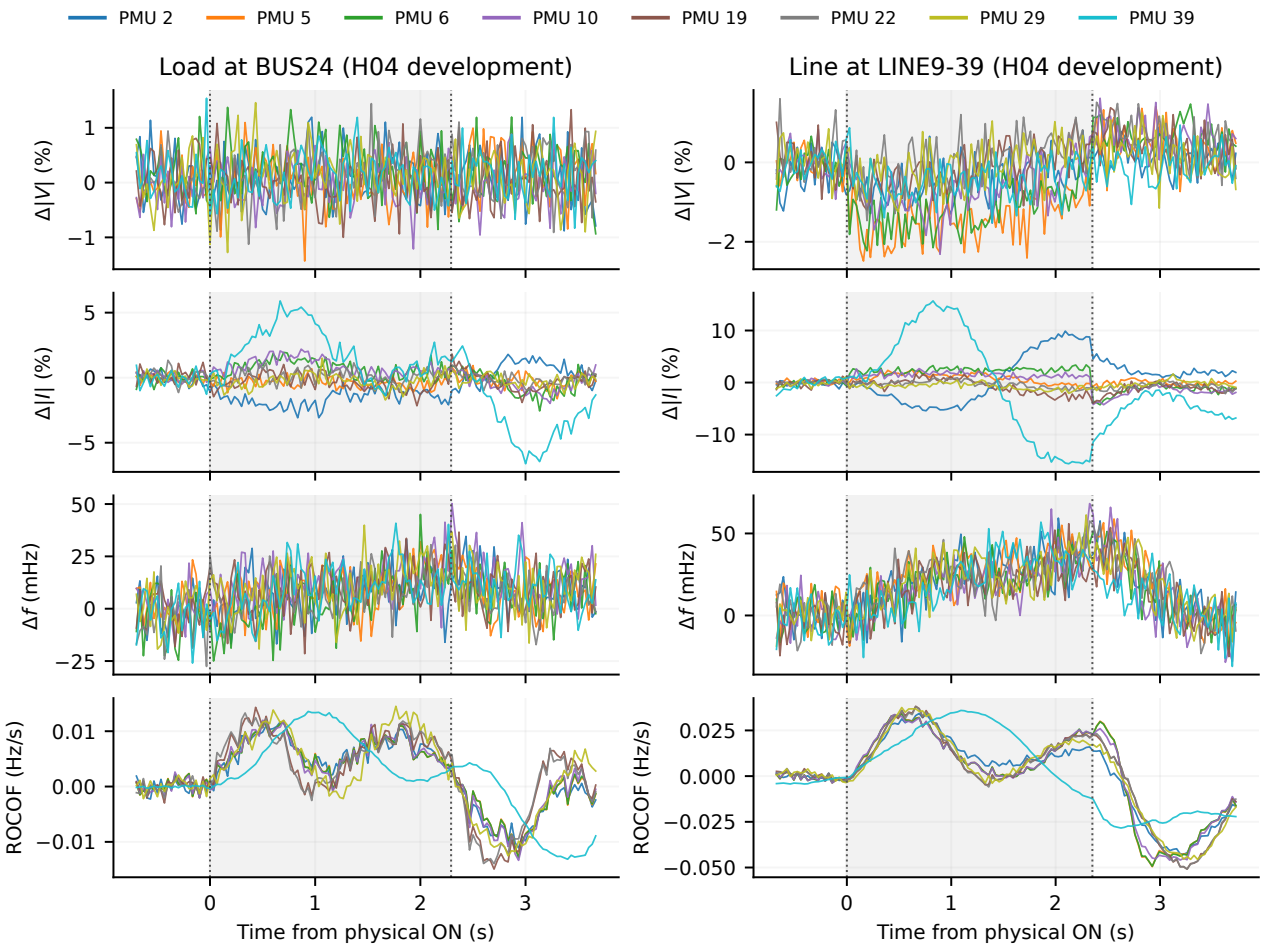


FIGURE 4. Simulated observations supplied to inference. One fixed development parent provides a load and a line example at all eight PMUs. Shading marks physical ON–OFF for evaluation only. Magnitude changes use the local pre-event center; frequency is centered for display while its exported channel remains absolute Hz. Source, severity and operating context affect the curves, so they are not universal class templates.

For conditional history mass $\pi_{\ell,h,k}$, define active and originating-cause masses for an admissible family/source cell c:

$$M_{c,k} = \sum_{\ell} w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}\, \mathbf{1}\{h \text{ active at } k\},$$
$$\Gamma_{c,k} = \sum_{\ell} w_{\ell,k} \sum_{h:c(h)=c} \pi_{\ell,h,k}. \tag{16}$$

A retained history can stop being active while its cause still explains recovery. Later interventions compete with those histories. The implementation stores 18 admissible cells, not every family at every bus. After confirmed episode closure, explicit histories are compressed into the retained state mean. Cause masses describe the current bank, not a permanent ledger of all past interventions.

Writing the conditional active mass as $M_{\ell,c,k}$ yields

$$\Delta M_{c,k} = \sum_{\ell} w_{\ell,k}(M_{\ell,c,k} - M_{\ell,c,k-1}) + \sum_{\ell}(w_{\ell,k} - w_{\ell,k-1})M_{\ell,c,k-1}. \tag{17}$$

The terms distinguish within-model evolution from model reweighting. This identity separates within-model evidence from network-model reweighting. The masses are approximate attribution weights; probability calibration and globally optimal filtering are not established.

F. EMIT SEPARATE DETECTION, DIAGNOSIS AND RESTORATION MESSAGES

The fixed publication rules require active mass 0.99 for six reports to confirm an alarm, and restoration mass 0.99 for 15 reports to confirm OFF. The archived source/line confirmation and episode-replacement rules determine the public diagnosis. Every message records emission time and the last report used. A diagnosis can be correct eventually but too late to precede physical OFF.

The nominal comparator uses the same nominal conditional model with its own fixed publication path. Comparing nominal and mixture evaluates two complete configured procedures; it does not isolate the outer network bank from publication decisions. The admission-rule ablation tests a different publication decision; it is not a pure recovery-memory

ablation.

The per-report birth and restoration hazards are $1 - \exp[-(1/60)/30]$ and $1 - \exp[-(1/30)/30]$. Birth mass is split equally between load and line families, then uniformly among sources within each family. Load severity has a zero-mean Gaussian prior with standard deviation 0.15, truncated to $[-0.2, 0.2]$ and integrated over four intervals with boundaries $-0.2, -0.1, 0, 0.1, 0.2$. After three reports, eligible active histories can produce OFF children at either half-report position while inheriting state and severity. After likelihood normalization, the beam retains the 96 highest-weight recent histories, 64 older histories and at least one per occupied source; their union is renormalized and discarded mass is recorded. The reproducibility archive maps these operations to the frozen `transitions`, `process`, and `predictive_log_density` routines and preserves the complete publication configuration.

G. RELATION TO MULTIPLE-HYPOTHESIS FILTERING

For a retained history h, let $p_\ell(h \mid h')$ contain the admitted birth, continuation and restoration transitions. Before pruning, its normalized evidence update has the standard form

$$\pi_{\ell,h,k} \propto L_{\ell,h,k} \sum_{h'} p_\ell(h \mid h') \pi_{\ell,h',k-1}. \tag{18}$$

IMM filtering approximates mode-conditioned state mixtures; the M3H filter retains mode-sequence hypotheses [20], [21]. Equation (18) is that established mechanism. Here the modes are oriented electrical intervention operators, histories include restoration time, and the output separates active intervention from its originating cause through (16). Retrospective births reuse a stored pre-onset state, while closure compresses histories into a retained augmented-state mean. This specifies an application of history filtering rather than a new general recursion. The state-carry experiment tests one operation in this construction; the preliminary single-state control below requires independent, information-matched validation to identify the incremental benefit of explicit recovery hypotheses.

H. PER-REPORT ALGORITHM AND COMPUTATIONAL BOUNDARY

Algorithm 1: causal physical-history attribution. The inputs are the nominal model, fixed method configuration and successive PMU reports. Reserved event truth never enters these steps.

1) Estimate operating context, common reference and noise scale from the startup prefix. Initialize conditional states and retain the eligible past states needed by retrospective proposals.
2) At report k, apply the frozen centering, scaling and instrumental transformation. Propagate each retained physical/recovery history and its conditional uncertainty to predict this report. Compute and save the outer model predictive density before current-report screening and state correction.
3) Apply the frozen short-lag birth screen using retained past states. A proposed history starts before its selected onset; replay replaces the corresponding likelihood suffix rather than adding it twice.
4) Score candidate histories and replayed suffixes before correcting their states with the reports being scored. Integrate the allowed load severity and apply the archived switched operator for a line hypothesis.
5) Correct the conditional states and severity distributions; update network weights and the retained history bank using the fixed rules. Preserve dynamic state and originating cause through restoration.
6) Aggregate active mass M, cause mass Γ and ΔM. Apply the fixed alarm, diagnosis and OFF-publication rules. Store emission time, estimated onset, family/source and the last report used.

The nonlinear simulator constructs observations; this algorithm propagates local switched predictors. The input grid is 30 reports/s. Causality concerns report availability; acquisition throughput, computation time and the electrical history needed for a decision are separate quantities. Wall-clock real-time operation has not been established.

V. EXPERIMENTAL DESIGN

The primary experiment asks whether the causal bank can attribute successive load/line interventions without confusing recovery with a new event. Development, independent evaluation and an auxiliary source-resolution experiment answer different questions and retain separate denominators.

The simulator is the IEEE-39 Julia/PowerDynamics research candidate. Each physical parent is integrated continuously, preserving differential states through ON/OFF. Its PMU reporting grid is separate from solver steps: a 120-s record contains 3601 reports per PMU and a 600-s normal record contains 18001, including endpoints. At a switch-aligned report the export uses the post-switch algebraic value. Measurement references, load law and stochastic instrument parameters are fixed across events. RAW0001 supplied retrospective calibration context, not independent test data.

The two development campaigns each contain three event parents with six successive load/line interventions and one normal parent, all 80 s long. Schedules are paired across nominal, in-bank R/X and mixed out-of-bank R/X conditions. Each campaign therefore has 18 unique interventions and 54 event realizations. Both campaigns are now development evidence. No interval or generalization claim treats those 54 realizations as independent events.

The five-model bank and its nominal version are frozen co-references. Both ExtraTrees artifacts and their causal preprocessing/publication adapters are frozen as well; neither is retrained on the development streams or the independent test. They receive the same 30-fps observations. This is a comparison of those operational implementations on the stated contract, not a rerun of the hackathon leaderboard.

The published physical comparator is Chen *et al.*'s persistent single-line quickest-change detector [2]. Its reduced-

TABLE 3. Candidate assets fixed during method development, before the independent schedules. This is a known catalog rather than a representative sample of all network assets.

Family	Candidate physical sources
Load buses	3, 4, 7, 15, 16, 20, 24, 26, 31
Lines	3–4, 4–14, 9–39, 14–15, 16–17, 17–18, 23–24, 26–27, 26–29

network covariance is marginalized to the observed PMU subset after inversion. Parallel CUSUM statistics stop at the first threshold crossing and select a line. Our declared research-plant adapter uses nonoverlapping angle increments at 15 Hz, a startup-only scale/drift estimate and the fixed nominal line catalog. Its white-injection/DC assumptions do not describe all simulated instrumental effects. We preserve both matched-law implementation checks and model-mismatch failures; no ARL guarantee is transferred to colored noise.

The eligible common test for Chen is one persistent line opening or a normal record. Its output has no load family or restoration operator. The independent campaign therefore separates continuous load/line episodes from persistent-line first-stop decisions. Early stops, wrong sources, omissions and horizon censoring remain visible.

For continuous events, an alarm is associated temporally before predicted family/source is inspected. The first unmatched alarm after ON is eligible up to the next ON, with the inherited 30-s cap. A timely joint decision requires the correct family and asset emitted before true OFF. Unmatched alarms are extras; false restorations and open episodes are reported separately. These definitions prevent an eventual correct answer from hiding a late diagnosis.

The independent campaign specifies 18 continuous parents, 27 persistent-line parents and 12 normal parents, with three R/X conditions and instrumental-noise scales of one and two. The 171 intended physical trajectories yield 342 observed records. Conditions represent nominal parameters, perturbations represented in the bank and mixed perturbations outside it. Solver and inference failures remain in the intended denominators. Comparisons use 10,000 paired parent-cluster resamples; repeated conditions/noise do not count as independent parents. Predictions are fixed before the evaluator opens reserved truth.

Logical decision delay is measured on report timestamps. Wall-clock cost is recorded separately with hardware, process concurrency and one-thread BLAS settings; parallel execution of independent records does not make a within-record inference method real-time. The independent results are followed by a development example and the controlled component study.

VI. RESULTS AND CONTROLLED COMPARISONS

A. INDEPENDENT EVALUATION: TIMELY ATTRIBUTION AND ALARM COST

The five-model bank makes 644/648 timely family-and-source decisions with 72 unmatched event-stream alarms. Its nominal counterpart makes 643/648 with 92 extras. The frozen causal ExtraTrees reference makes 330/648 with 29; its short-window variant makes 352/648 with 16 extras. These are paired replays of 108 interventions, not 648 independent events. Figure 5 shows why the totals differ: the physical banks recover nearly all loads, while both learned adapters already recover most tested lines. The extra alarm burden of the physical banks rises sharply when true R/X changes fall outside their five-model bank.

Figure 6 separates a fast conditional median from coverage: a method can have short delays on its available diagnoses while leaving many load events without a timely source decision.

A posthoc audit of the original episode matches further separates coverage from classification. Every matched diagnosis emitted before OFF has the correct family and source in all four procedures; their errors in the timely endpoint are missing or late diagnoses. The load advantage therefore concerns timely coverage. It does not demonstrate higher conditional classification accuracy among diagnoses already available before OFF. The supplement shows the complete confusion view and counts unmatched alarms separately.

The paired intervals in Fig. 7 confirm a timely-attribution advantage over this ExtraTrees reference in each tested stratum. They also show a higher rate of unmatched alarms under out-of-bank R/X. Those intervals resample physical parents, preserving correlation across noise replays and network conditions. They do not establish superiority at a common false-alarm operating point. The physical bank's stratum median diagnosis delay is about 0.27 s in matched and in-bank conditions; the ExtraTrees medians range from 0.19 to 0.30 s. There is no general latency advantage.

The separate persistent-line test scores the first stop. Both physical banks and both ExtraTrees adapters identify the right line first on 162/162 openings. The declared Chen2016 adapter does so on 74/162; 66 records stop before the opening and 22 stop afterward on a wrong line. A separate descriptive audit of the same persistent records shows 164 and 179 later unmatched alarms for the two ExtraTrees procedures, versus 0 for the physical bank. Correct first identification therefore does not imply stable maintenance of a sustained event. This audit does not change the first-stop comparison or apply a recurrent score to Chen. Chen also stops on all 72 paired normal records, whereas the continuous methods alarm on none. Chen's normal exposure ends at its first stop (2,522 s); the other methods monitor 42,336 s each. These unequal exposures and the adapter's white-injection assumption prevent a general claim against published quickest-change detection. Full stratified tables, failure ledgers, restoration outcomes and censoring are in the supplement.

The physical dictionary supplies responses for each catalog source, whereas the learned adapters retain their original

TABLE 4. Experimental populations and their purposes. Conditions and noise realizations repeat a physical parent rather than creating independent trajectories.

Population	Configuration	Main outcome	Interpretation
Continuous development	Three event parents, six interventions each; one normal parent; three network conditions	Timely family/source, extras and restoration	Open development evidence, not independent confirmation
Independent continuous evaluation	18 event parents, six interventions each; 12 normal parents; three network conditions and two noise scales	Paired attribution, false alarms, delays and failures	Prespecified comparison of unchanged methods
Persistent-line evaluation	Nine lines, three parent repetitions; same network/noise strata	First stopping decision, source and censoring	Eligible comparison with published line-outage QCD
Supplementary signature test	57 dictionary and 85 test trajectories; four loads; three common noise seeds	Nominal, reference correction and mixed-signature correction	Known ON/OFF and family; separate from causal accuracy
Supplementary recovery geometry	38 trajectories; four recovery histories and 12 correlated source-pair comparisons	Separation with known history and one uncertain severity direction	Supplied prior trajectory, onset, family and pair; separate from stream accuracy

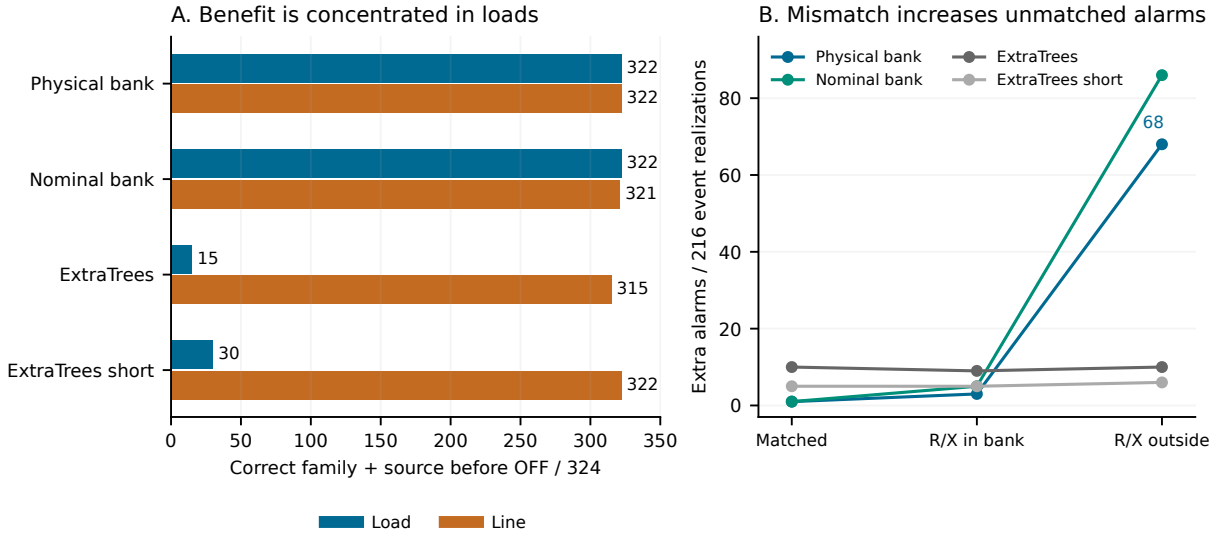


FIGURE 5. Sealed continuous evaluation. (A) Timely correct family and source by event family, with 324 paired realizations per family. (B) Unmatched event-stream alarms across the two noise levels within each R/X condition, with 216 event realizations per condition. The plots display the gain in load attribution and the alarm cost under out-of-bank network error together. Each method retains its own frozen publication rule and false-alarm budget.

TABLE 5. Sealed continuous-stream decisions. Each family has 324 replayed instances from 54 distinct interventions (108 across both families); extras are unmatched alarms on event records. The methods retain unequal alarm operating points.

Method	Load	Line	Extras
Physical bank	322/324	322/324	72
Nominal bank	322/324	321/324	92
ExtraTrees	15/324	315/324	29
ExtraTrees short	30/324	322/324	16

training and feature maps. The comparison therefore measures transfer of those particular implementations, rather than an information-matched test of physics against supervised learning. It also uses different alarm operating points.

TABLE 6. Recorded stage times from the continuous evaluation on 120-s records. Medians and ranges summarize full-record stages; the per-report column is a median-total divided by 3,601 reports and is amortized, not frame latency. Stages were recorded in a campaign with one BLAS and one Julia thread; memory and per-report latency were not recorded. This descriptive table does not establish real-time operation or a speedup under matched implementation conditions.

Stage	Median (s)	Range (s)	ms/report†
Bank build	80.1	62.0–309.7	22.2
History inference	358.7	218.2–2641.1	99.6
ExtraTrees (both)	9.5	6.7–50.5	2.6
Chen detector	1.3	0.9–10.6	0.4

B. COMPUTATIONAL COST

The recorded median history-inference stage exceeds the record duration; its amortized processing rate is below the

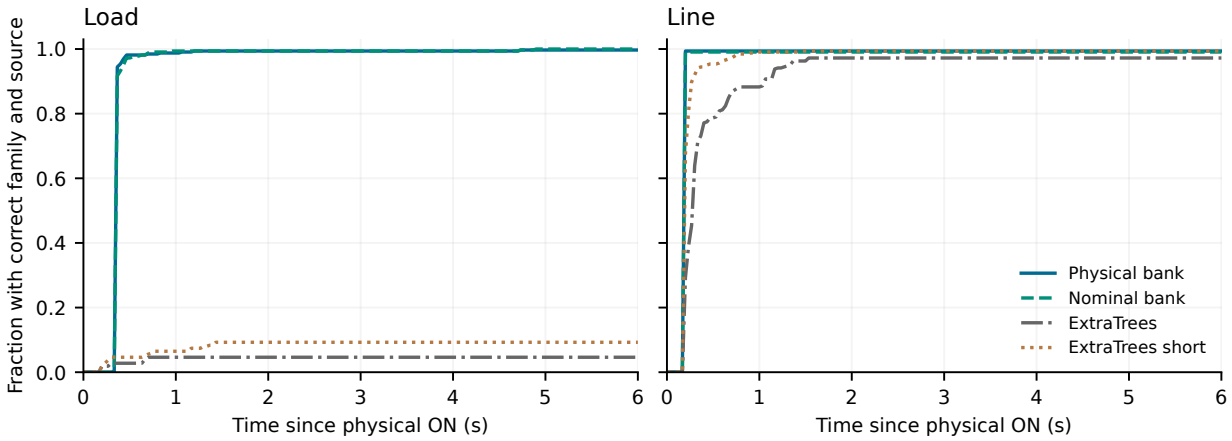


FIGURE 6. **Posthoc descriptive distribution of correct diagnosis time, using all 324 realizations per true family in each denominator. An incorrect or absent diagnosis contributes no cumulative mass. Curves include late correct decisions and are therefore distinct from the before-OFF endpoint; physical OFF varies between events. The correlated replays are not independent observations.**

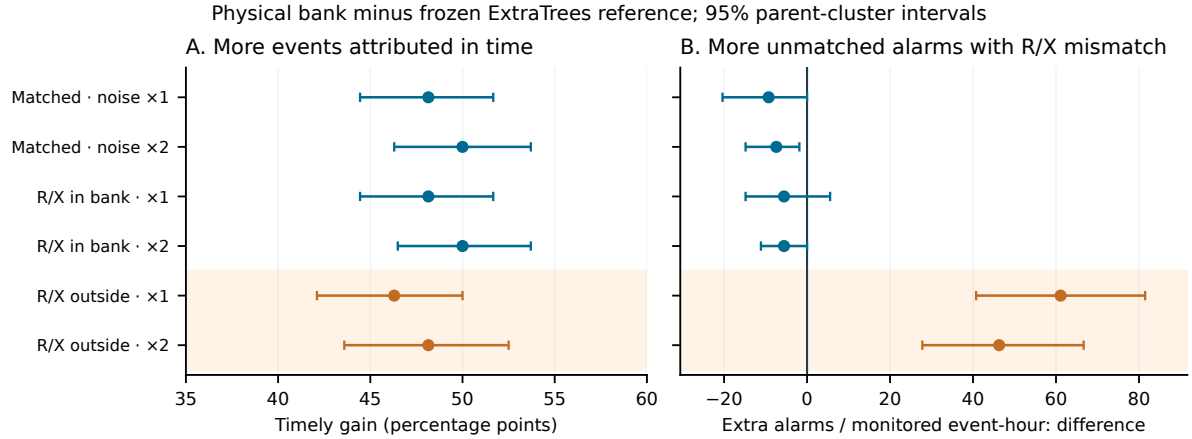


FIGURE 7. **Physical bank minus frozen ExtraTrees reference in the six paired strata. Points and 95% intervals come from 10,000 physical-parent resamples. Left: percentage-point difference in timely attribution; right: difference in unmatched alarms per monitored event-record hour, excluding the 12-s startup from each 120-s record. The original rate differences and interval bounds are rescaled exactly by 120/108; counts and episode matches are unchanged. The orange rows expose the out-of-bank R/X failure. The intervals describe this finite simulated population.**

30-report/s input cadence. The stage ranges show substantial variation. Per-report latency percentiles, peak memory and a matched-hardware speed comparison were not recorded; causal emission timestamps must not be read as wall-clock latency.

C. A CONTINUOUS SEQUENCE AND A DELAYED DECISION

A fixed development trajectory illustrates the timing criterion (Fig. 8). The network continues moving after OFF; a later event must be evaluated against that predicted response. Under the tested out-of-bank network error, the bus-31 source is recovered only after restoration. It is correct eventually and fails timely attribution. The full development counts, active-mass trace, ΔM decomposition and failed admission control remain in the supplement.

D. A CONTINUOUS SINGLE-STATE PHYSICAL CONTROL

An existing open-development control retains one committed augmented physical state through restoration, rather than resetting to equilibrium. It uses the same nominal conditional plant and measurement operators, profiles severity over short alternatives lasting at most 19 reports, and applies the same six-report publication clock to all three procedures. The short GLR achieves 37/54 timely decisions with 37 extras, versus 44/54 and 6 for the nominal history bank, and 51/54 and 3 for the five-model bank. These 54 replays share three development parents; they are separate from the independent test. Different evidence scores, search and severity updates remain, so this is a preliminary continuous-observer comparison, not an isolated causal effect of history retention. Its complete equations and closure outcomes are in the supplement.

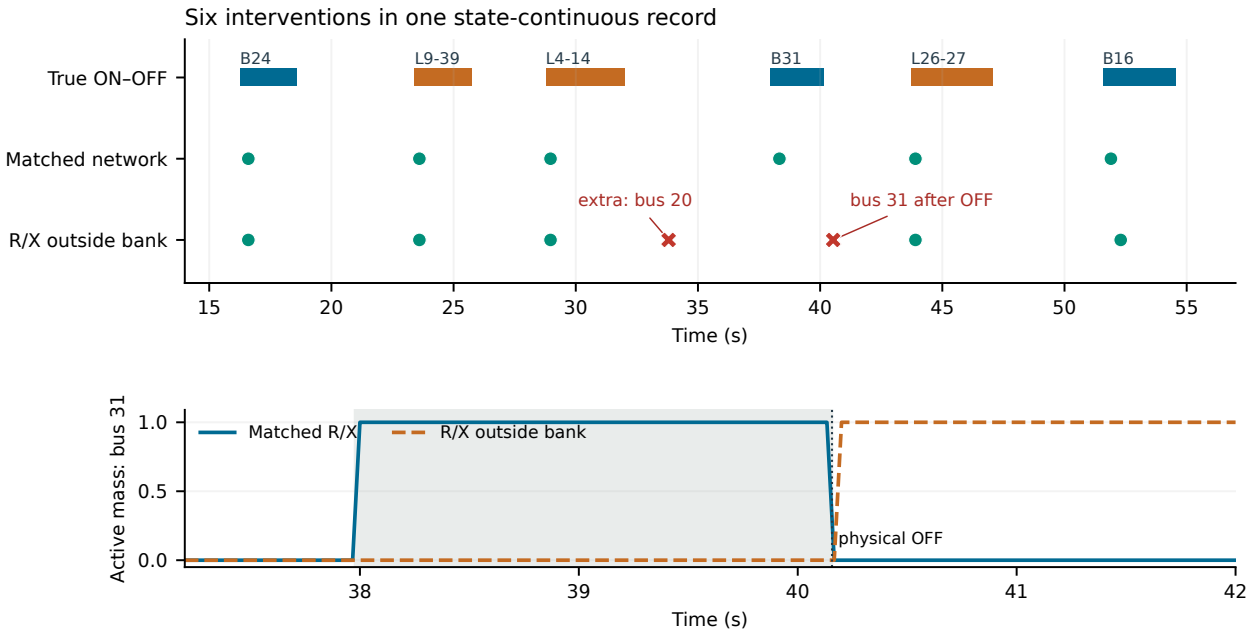


FIGURE 8. **One continuous development trajectory, shown for interpretation rather than selected as an independent test. Top: physical ON–OFF intervals and emitted diagnoses under matched and out-of-bank R/X. Bottom: bus-31 active mass, with the true intervention shaded. A diagnosis after OFF misses the timely endpoint even when it identifies the source. True timing enters only this retrospective display.**

E. CONTROLLED SEVERITY-SCORE AND STATE-CARRY STUDY

The continuous comparison evaluates complete procedures. To distinguish two mechanisms within the physical bank, we fixed a subsequent 2×2 study: marginal amplitude evidence versus a bounded profile score, and physical-state carry versus removal at inferred closure. The five network models, observations, startup fits, operators, proposal rules and publication thresholds are common to all four arms. Identical thresholds do not imply identical false-alarm rates; both coverage and alarm counts must be measured.

For one load history and amplitude interval $[l, u]$, write its accumulated conditional log likelihood as

$$\ell_k(a) = c_k + b_k a - \tfrac{1}{2} G_k a^2. \tag{19}$$

The original score uses $B_k = \log \int_l^u \exp(\ell_k(a) - c_k) p(a)\, da$ with the existing truncated Gaussian prior. The alternative uses

$$\widehat{a}_k = \mathrm{clip}_{[l,u]}(b_k/G_k), \qquad Q_k = b_k \widehat{a}_k - \tfrac{1}{2} G_k \widehat{a}_k^2, \tag{20}$$

with the constant-likelihood case treated separately. Each report adds $B_k - B_{k-1}$ or $Q_k - Q_{k-1}$, so a rolling replay does not repeatedly add the same suffix evidence. The conditional Gaussian amplitude/state updates remain unchanged. Profile weights are normalized scores, not posterior probabilities. This intervention tests the amplitude-scoring choice, rather than removing every Bayesian calculation. The existing Gaussian sufficient statistics supply $G_k = v_k^{-1} - v_0^{-1}$ and $b_k = \mu_k/v_k$, removing the computational prior precision before profiling. Fixed line-opening amplitudes and the prior weights of amplitude intervals are unchanged.

At an internally inferred closure, the carry arm retains the projected augmented state. The reset arm sets its 114 estimated physical-state deviations to zero and retains instrumental/discrepancy coordinates, covariance regime and network-model weights. This is an estimator ablation: every observed record comes from the same continuous physical trajectory. It does not test removal of all temporal information.

The protocol uses all 18 matched-network continuous parents at the original noise level, plus the first three normal and persistent-line parents by identifier. It was fixed before ablation outcomes, but the underlying records had already been examined. Results are therefore a posthoc mechanism study, with paired intervals over physical parents, not a new independent confirmation.

1) Component Outcomes

Table 7 and Fig. 9 report coverage and alarm cost for the four arms. With carry retained, marginal and bounded-profile amplitude scores tie on the reported endpoints. Resetting the first 114 estimated physical-state deviations at inferred closure reduces timely attribution and increases extras under both scores. This reset leaves instrumental state, covariance regime, prior weights, thresholds, clocks and reporting rules intact; simulated states and observations never reset. The profile arm remains a normalized penalized score system and retains conditional Bayesian state estimation. Thus this study supports a physical-state-carry effect at inferred closure in the tested cohort, not the necessity of Bayes or of every form of temporal memory. The supplement reports denominators, normal exposure, persistent episodes, censoring and all paired parent-cluster intervals. These are posthoc within-simulator

TABLE 7. Controlled amplitude-score and physical-state-carry ablation. Same five-model bank and publication rules. All 18 continuous parents (108 events) are retained. Normal controls comprise three 600-s records with 588 s evaluated exposure each (1,764 s total; startup excluded). Delay is conditional on emitted matched diagnoses and must be read with the no-timely-diagnosis count.

Procedure	Timely	Extras	Normal FA	False restorations	No timely	Delay (s)
Marginal + carry	108/108	1	0/1,764 s	1	0/108	0.267
Profile + carry	108/108	1	0/1,764 s	1	0/108	0.267
Marginal + reset	75/108	166	0/1,764 s	130	33/108	0.337
Profile + reset	75/108	166	0/1,764 s	130	33/108	0.337

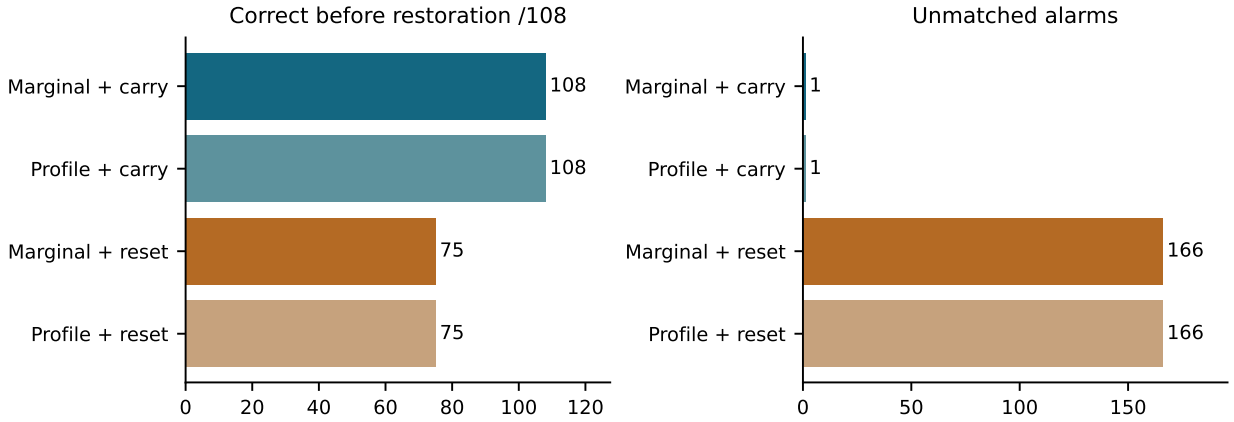


FIGURE 9. Amplitude-scoring and physical-state-carry ablation on identical observations and five-model banks. The 108 interventions belong to 18 continuous trajectories. Thresholds are fixed, so the observed alarm rate may differ between arms.

contrasts at the original thresholds, not a comparison at matched false-alarm rates.

VII. DISCUSSION AND LIMITS

A. PHYSICAL-STATE CONTINUITY AS A DESIGN CHOICE

The strongest controlled result is the state-carry contrast. Physical OFF restores an input or branch, while the machines retain their state. Removing those estimated deviations at inferred closure substantially worsens decisions on the same records. Marginal and bounded-profile severity scores tie on the measured endpoints; both retain conditional state estimation. This supports recovery-consistent prediction, without showing that Bayesian marginalization is necessary for those endpoints.

The history bank gives a concrete way to represent the remaining response and compare it with later interventions. Its complete operational behavior is measured, but the reset control is deliberately severe. A continuous physical observer without explicit recovery-history competition could retain the same consistency. The open-development single-state GLR comparison gives preliminary evidence against one such alternative, with different search and score semantics. Independent validation under matched information and a common calibration contract remains the key unresolved mechanism test.

B. WHAT IMPROVED RELATIVE TO THE LEARNED REFERENCE

The improvement concerns timely coverage of load sources under this known-catalog simulator contract. Line coverage is comparable to the short-window learned adapter. The physical method also issues more extra alarms under network mismatch. The result is therefore a coverage–alarm trade-off at fixed thresholds, rather than dominance at equal alarm rates or a higher official hackathon score. Engineering physical features is useful in the historical learned route; the physical bank additionally computes how each candidate intervention should affect remote PMUs. These routes have different access to source responses and training data.

A fairer architecture comparison would give a physical GLR or M3H control and a learned baseline the same candidate catalog and permitted development records, then select their operating points on validation data before a new test. Neither a posthoc threshold sweep on this test nor a new GNN name would establish that result. The present comparison preserves the original adapters and all their failures.

C. MODEL CORRESPONDENCE AND MEASUREMENT SCOPE

The nonlinear generator and local predictors share the same research plant family. Switched dynamics, nonlinear responses, noise and out-of-bank R/X challenge the inverse model, but do not remove this shared-model advantage. Independent dynamic-model or cross-simulator transfer remains untested. All nine load and nine line candidates were inherited from method development before the independent draw; they are a declared catalog, not a representative sample of all 39 buses or all event families.

The eight-PMU terminal map is a research measurement hypothesis. BUS5 retains the provisional 5–6 assignment,

and BUS39 current angle is excluded. The generator applies temporal and cross-PMU noise structure within each signal family, but does not apply its archived cross-family factor covariance. Frequency/ROCOF noise is added after phase estimation. The experiment therefore does not validate a complete physical PMU noise model, a producer's CT placement or exact reproduction of the historical reference record. It also assumes an event-free startup and available PMUs.

D. APPROXIMATION AND DECISION UNCERTAINTY

History pruning, shared regime covariances, plug-in startup estimates and state-mean compression make the bank approximate. Its masses are attribution weights under those approximations, not validated confidence levels. Discarded mass is recorded by the implementation; a calibrated reliability study and a bank-size sensitivity study are still missing. Response fidelity and source discrimination also differ: sources may remain close after fitting severity and context. The supplementary context-signature and recovery-geometry controls preserve their null source-decision results and do not supply an online identifiability certificate. A useful extension would test whether a causal separation margin predicts errors and delays before using it to justify abstention.

The measured research implementation runs below the input cadence. Decision timestamps, computation delay and queueing must remain separate. These results support further diagnostic development, rather than protection or control deployment.

VIII. CONCLUSION

A restored electrical intervention can remain the cause of the observed response. We formulated load/line attribution through physical histories that start from estimated past states and retain that response through restoration. Active and originating-cause weights separate the current intervention from its continuing physical effect.

The independent simulated comparison gives 644/648 timely family/source decisions and 72 extra alarms for the five-model bank, compared with 330/648 and 29 for the fixed causal ExtraTrees reference. The advantage is concentrated in loads; line coverage is similar to the short-window learned comparator. The controlled study identifies a strong state-carry effect, while marginal and profiled severity scores tie. Together, these results support an executable approach that uses electrical dynamics to attribute interventions from remote PMUs, preserves the response across restoration, and separates timely coverage from alarm burden. Physical-state continuity supplies the consistent reference needed to evaluate subsequent interventions during recovery.

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